# Survival Analysis of Rats Data

## A Comparative Study of Parametric and Regression Models

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github.com/walternyamutamba-svg/rats-survival-analysis

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# Objective

## Goal

Fit six parametric survival distributions to the `rats` tumour-incidence data, select the best model using AIC, and derive the four life functions: $S(t)$, $h(t)$, $f(t)$, $H(t)$.

## Why Parametric Models?

- Kaplan-Meier is non-parametric which means it has no closed-form expression
- Parametric models yield analytic formulas for survival, hazard, density, and cumulative hazard
- Enable direct computation of mean, median, and variance of survival time
- Allow extrapolation beyond the observed follow-up period

# Dataset & Study Design

**Source:** `rats` dataset, `survival` package in R

**Variables used:**

- `time` — days to tumour or censoring
- `event` — 1 = tumour, 0 = censored
- `treatment` — Control vs Treated (rx)
- `sex` — female / male covariate
- `litter` — clustering variable (100 litters)

**Design:** 2:1 allocation
200 Control, 100 Treated
Max follow-up = 104 days

*Treated rats received a carcinogenic agent (rx = 1)*

## Sample Summary

| Quantity | Value |
|---|---|
| Total rats | 300 |
| Events (tumours) | 42 |
| Censored | 258 |
| Overall event rate | 14% |
| Control events | 21/200 (10.5%) |
| Treated events | 21/100 (21.0%) |
| Mean survival | 90.4 days |
| Missing values | None |

**Note:** Low event rate (14%) means median survival is not estimable so we report RMST instead.

# Descriptive Overview

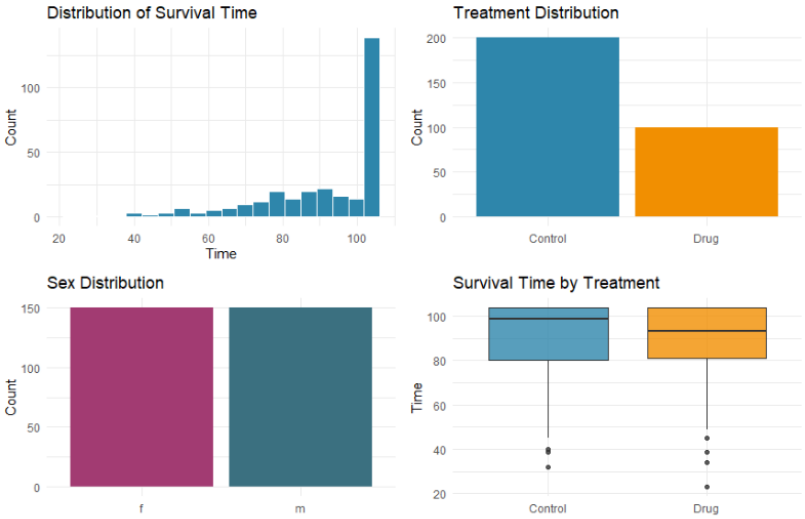


Figure 1: Distribution of survival times and comparison by treatment group

# Kaplan–Meier Survival Curves

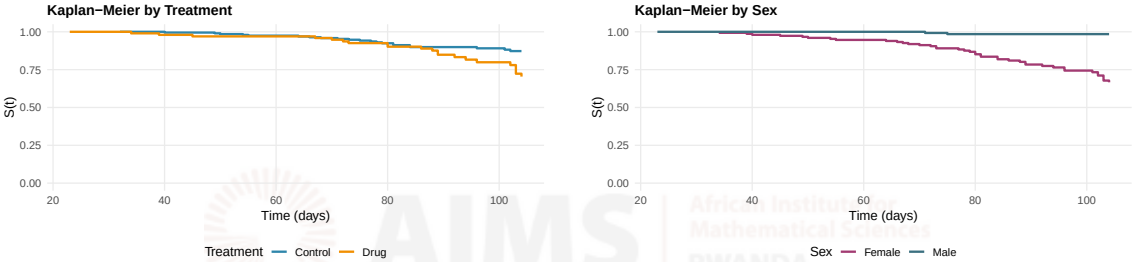


Figure 2: Kaplan–Meier survival curves

## Interpretation

- Females experience substantially more tumour events than males (40 vs 2).
- The treatment effect appears modest, with similar survival patterns between groups.
- Kaplan–Meier curves provide a non-parametric reference for evaluating parametric survival models.

# Parametric Model Fitting and Selection

Table 1: Parametric Model Values

| Model | Params | Estimates | AIC | loglikelihood |
|---|---|---|---|---|
| Exp | $\lambda$ | 0.00155 | 629.54 | -313.77 |
| Weibull | $k, \lambda$ | 3.67,160.67 | **578.20** | -287.10 |
| LogN | $\mu, \sigma$ | 5.16,0.56 | 578.86 | -287.43 |
| LL | $\alpha, \beta$ | 153.95,3.82 | 578.32 | -287.16 |
| Gamma | $k, \lambda$ | 5.48,0.032 | 578.38 | -287.19 |
| Gomp | $b, \eta$ | 0.0001,0.044 | 580.66 | -288.33 |

## Why AIC?

- Balances goodness of fit and model complexity
- Penalises models with too many parameters
- Used to compare non-nested survival models
- Lower AIC indicates a better model

# Comparison — All Models vs KM

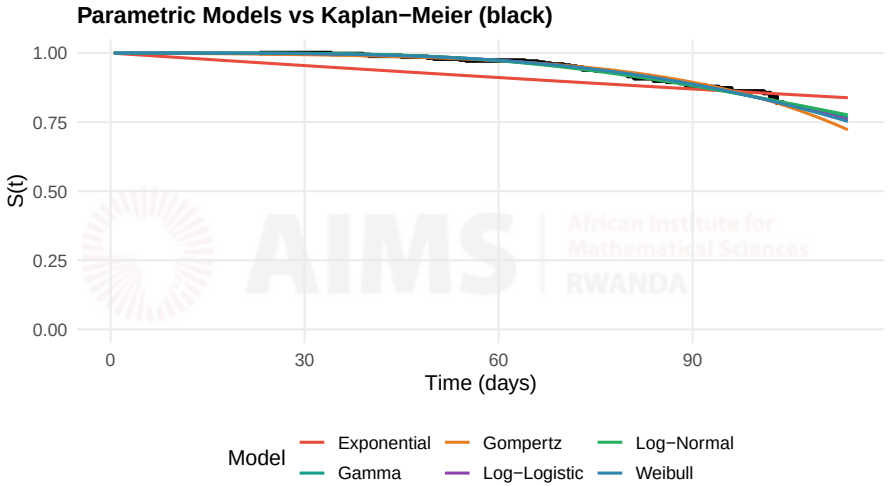


Figure 3: Comparison of parametric survival models with the Kaplan–Meier curve (black).

# Why Weibull?

## Statistical Justification

- Lowest AIC (578.2) among all six candidates
- Closest visual alignment with non-parametric KM curve
- Includes Exponential as a special case ($k = 1$), so rejecting Exponential supports $k \neq 1$

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## Practical Justification

- Shape parameter $k$ directly characterises the hazard trend
- Closed-form expressions for all four life functions
- Mean, median, and variance available analytically
- Widely used and interpretable in biomedical research

# Weibull Life Functions

With shape $k = 3.67$ and scale $\lambda = 160.67$

**Survival function**

$$S(t) = \exp\left[-\left(\frac{t}{\lambda}\right)^k\right]$$

**Hazard function**

$$h(t) = \frac{k}{\lambda}\left(\frac{t}{\lambda}\right)^{k-1}$$

**Density function**

$$f(t) = h(t) \cdot S(t)$$

**Cumulative hazard**

$$H(t) = \left(\frac{t}{\lambda}\right)^k$$

Since $k = 3.67 > 1$, the hazard $h(t)$ is monotonically increasing which means tumour risk accelerates over time.

# Weibull Life Functions

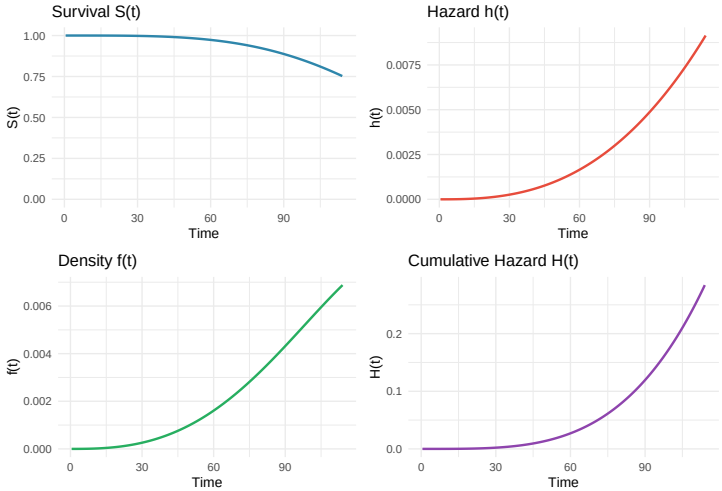


Figure 4: Weibull life functions showing survival $S(t)$, hazard $h(t)$, density $f(t)$, and cumulative hazard $H(t)$.

# Interpretation

## Parameter Meanings

- $k = 3.67$ hazard is increasing and convex which means tumour risk accelerates with age
- $\lambda = 160.67$ the characteristic life (time at which $S(t) = e^{-1} \approx 0.37$)

## Summary Statistics

Table 2: Summary Statistics

| **Quantity** | **Value** |
|---|---|
| Mean survival | 144.9 days |
| Median survival | 145.4 days |
| Variance | 1 934.8 days$^2$ |
| Std. deviation | 44.0 days |

# Conclusions

1. The Weibull distribution provides the best fit (AIC $= 578.2$), closely followed by Log-Logistic and Gamma
2. The shape $k = 3.67 > 1$ confirms an increasing hazard which meant that tumour risk grows with time
3. Mean tumour-free survival is approximately 145 days
4. The Exponential model (constant hazard) is decisively rejected ($\Delta$AIC $= 51.3$), justifying the need for a flexible parametric model

## Limitations

- Rats from the same litter may be similar. A shared frailty model can account for this clustering.
- Treatment and sex were not included in the parametric models.

# References


Mantel, N., Bohidar, N. R., & Ciminera, J. L. (1977).
Mantel-Haenszel analyses of litter-matched time-to-response data, with modifications for recovery of interlitter information.
*Cancer Research*, 37(11), 3863–3868.

Kleinbaum, D. G., & Klein, M. (2012).
*Survival Analysis: A Self-Learning Text* (3rd ed.).
Springer.

Therneau, T. M. (2023).
*A Package for Survival Analysis in R*.
R package version 3.x.

Collett, D. (2015).
*Modelling Survival Data in Medical Research* (3rd ed.).
CRC Press.


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